

Galactic-scale validation of the BPB/MBE MAP1782 branch with SPARC rotation curves

Paper II draft v0.4.1 - corpus-aware galactic response theory and SPARC validation

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Abstract

Paper I froze a low-redshift plus KiDS plus ACT working point of the BPB/MBE framework without using SPARC rotation curves. This Paper-II draft is deliberately self-contained and corpus-aware: it first formulates a minimal galactic response theory for how the frozen MAP1782 spectrum should enter disk rotation curves, and only then tests that response on SPARC. The theoretical point is that a rotation curve is not a direct mass-variance observable. It is sensitive to a radial or concentration-like response, so the natural first bridge is not a smoothed W^2 mass kernel but a logarithmic-gradient kernel, whose leading proxy is kW^2 , coupled to the measured outer rotation-curve slope m_{out} . The fixed pre-test sign rule is therefore $-\text{sign}(\Delta_{kW^2} m_{\text{out}})$ on the frozen active SPARC subset of 38 galaxies. It gives 33/38 correct signs, $p_{\text{perm}} = 0.00014$, and $Z_{\text{perm}} = 4.059$ under target-sign permutations. A naive W^2 kernel collapses to the morphology-only baseline and misses the high-mass positive response tail; kW^2 recovers that tail, including the three decisive galaxies F571-8, ESO079-G014, and ESO563-G021. When inserted into the validated archived rotation-curve machinery, the same 33/38 rule gives $\Delta\chi^2 = -4605.230$, within $\Delta\chi^2 = 8.340$ of the archived oracle value -4613.570 and improving the older physical archive rule by 497.640 in χ^2 . The v0.4.1 source-lock keeps the D5/D6 amplitude layer: a source-only row-dependent kernel, validated directly in raw point-level SPARC rotation curves, gives $\Delta\chi^2 = -4302.456$, recovers 0.773245 of the clean-to-archive amplitude gap, and survives a 5000-draw source-feature permutation null with empirical $p = 1/5001$. D3 supplies the density-weighted Weyl-matter projector $r_{W\delta} = \sqrt{a_{\text{bg}}} = 0.635364$, while D7 verifies the associated face-minus tilt mechanism in CAMB; both are guardrailed as supporting corpus locks rather than replacements for a full galaxy-formation or full Planck likelihood derivation. A complete nonlinear baryon+halo derivation of the derivative kernel remains the principal open theoretical task.

Keywords: galaxies: kinematics and dynamics - galaxies: spiral - dark matter - cosmology: theory - methods: statistical

1 Paper-II scope and Paper-I anchor

Paper I established the current low-redshift reference point of the BPB/MBE programme. The numerical anchor used here is fixed before any SPARC test: the BPB/MBE comparator has

$$\chi_{\text{BPB}}^2 = 1782.442077, \quad \chi_{\Lambda\text{CDM}}^2 = 1798.271219, \quad (1)$$

so that $\Delta\chi^2 = -15.829142$ for the Paper-I joint data vector. The same working point has $s_{\text{surv},0} = 0.7956924397708606$ in the exact YAML anchor, $\beta_{\text{grow}} = -0.2106621406010265$, and $c_{\Sigma} = 0.71$ for the ACT stress-gate closure. The central Paper-I claim is not that one scalar modifies all observables through a single window; it is that $\mu_m \neq \Sigma$, and that the Paper-I improvement is driven by the background/growth/KiDS sector while ACT acts as a Weyl stress gate.

This paper deliberately uses that asymmetry. SPARC is not added to the Paper-I likelihood. Instead, it is used as an external validation target: given the already frozen MAP1782 branch, does the reconstructed galactic response predict anything non-trivial in SPARC?

The answer supported by the present artefacts is conditional but sharp:

The MAP1782 spectrum predicts the active-SPARC correction sign only after it is filtered through a derivative-like response kernel, well approximated by kW^2 , and coupled to the observed outer rotation-curve slope m_{out} .

The safe claim is therefore a bridge claim, not a full galaxy-formation claim. We do not claim that BPB/MBE predicts m_{out} from first principles in this draft, and we do not claim that a raw, uncalibrated kW^2 amplitude is already the final rotation-curve model. We claim that the sign structure and the rotation-curve half-space selected by $\text{MAP1782} + kW^2 \times m_{\text{out}}$ are statistically and dynamically non-trivial.

2 Corpus placement and release navigation

This manuscript is Paper II of the BPB/MBE corpus. It should be read after Paper 0, which defines the BPB/MBE dictionary and the density-weighted branch quantities, and after Paper I, which freezes the low-redshift MAP point used here. The release package also contains four root navigation files: `00_README_CORPUS`, `00_MANIFEST_CORPUS`, `00_CLAIMS_INDEX`, and `00_REVIEWER_ROUTE`. These files are not additional physics assumptions. They are the audit layer that maps each paper claim to its source-lock artefact.

The role of Paper II in the corpus is specific. It tests whether the independently frozen MAP branch leaves a non-trivial, falsifiable signature in SPARC rotation curves. In the final corpus dashboard, the Paper-II chain contains five locks:

1. D3: a density-weighted Weyl–matter projector, with $r_{W\delta} = \sqrt{a_{\text{bg}}}$;
2. D3b: an s_0 -weighted HR scalar-projector diagnostic that predicts a_{bg} at sub-percent level but is not exact;
3. D5: a raw point-level SPARC validation of a row-dependent source-only amplitude kernel;
4. D6: the corresponding source-feature permutation, top-driver removal, and feature-ablation falsification layer;
5. D7: a CAMB face-minus tilt mechanism check, kept as a mechanism gate and not promoted to a full Planck likelihood.

The SPARC-specific heart of the paper remains the D5/D6 galactic response test. D3 and D7 are included to make the corpus dependencies explicit, not to blur the distinction between projected Weyl diagnostics, CMB spectra, and raw galaxy rotation curves.

3 Projector and mechanism locks used by this paper

Two corpus locks sit upstream or adjacent to the SPARC kernel test. They are recorded here because they control how Paper II should be read, but neither one is allowed to replace the raw SPARC validation.

First, D3 closes the projected Weyl–matter normalization as a density-weighted two-sector projector. With sector weights a_{bg} and $1 - a_{\text{bg}}$, the mixing angle satisfies

$$\tan^2 \theta_W = \frac{1 - a_{\text{bg}}}{a_{\text{bg}}}, \quad \cos^2 \theta_W = a_{\text{bg}}, \quad r_{W\delta} = \sqrt{a_{\text{bg}}} = 0.635364421415. \quad (2)$$

The diagnostic support is independent of defining the projector: the T8 amplitude-space check gives $\langle X, X \rangle / \langle A, A \rangle = 0.397990327336$, differing from $a_{\text{bg}} = 0.403687948000$ by -1.411392% , and fixing $r_{W\delta} = \sqrt{a_{\text{bg}}}$ changes the T1 diagnostic by only $\Delta\chi^2 = 0.008545$. The D3b s_0 -weighted condition

$$\frac{y_t^2}{s_0} (1 - a) = a^3 \quad (3)$$

predicts $a = 0.405837814272$ with rounded $s_0 = 0.800$, a 0.532556% offset from the locked value, and $a = 0.406433034785$ with exact $s_0 = 0.7956924397708606$, a 0.680002% offset. This is used only as a sub-percent mechanism diagnostic. It is not claimed as an exact β_n -only identity.

Second, D7 verifies the face-minus tilt mechanism at CAMB spectra level. The fixed displacement is $n_s = 0.972 \rightarrow 0.982$, i.e. $\Delta n_s = 0.010$, on the BPB CMB background. Over $\ell = 600\text{--}2200$, the same-background effective slopes are 0.008945 in TT and 0.009453 in EE. This supports the mechanism that connects the face-minus branch to a high- ℓ tilt-like response. It is not a full Planck likelihood and is not used to tune the SPARC rotation curves.

4 Minimal galactic response theory

This section is placed before the SPARC audit on purpose. The $kW^2 \times m_{\text{out}}$ rule is not introduced as a posterior rescue of the sign table; it is the minimal response rule expected if the Paper-I spectrum is observed through a galactic rotation-curve window.

4.1 Observable windows are probe-dependent

Paper I already requires a distinction between the matter-growth response μ_m and the Weyl response Σ . The same logic applies at galactic scales. A disk rotation curve does not measure $\mu_m(a)$ directly, nor the CMB-lensing auto-response Σ_{auto} . It measures a quasi-static radial force profile after the frozen cosmological response has been projected through a halo and disk state. We therefore write schematically

$$\mathcal{O}_{\text{gal}} = \mathcal{W}_{\text{gal}} [\mathcal{R}_{\text{MAP}}(k), H_{\text{ref}}, \mathcal{S}_{\text{disk}}], \quad (4)$$

where $\mathcal{R}_{\text{MAP}}(k)$ is fixed by Paper I, H_{ref} is the reference NFW halo used in the SPARC audit, and $\mathcal{S}_{\text{disk}}$ is an observed state variable of the galaxy. In this draft, the only disk-state variable entering the fixed sign rule is the outer slope m_{out} .

4.2 From spectral response to halo response

Let

$$\mathcal{R}_{\text{MAP}}(k) \equiv \frac{P_{\text{BPB}}(k)}{P_{\text{ACDM}}(k)} - 1 \quad (5)$$

be the frozen spectral response associated with the MAP1782 branch. A mass-amplitude response at smoothing scale R would be captured by a variance-like functional

$$\Delta_{W^2}(R) = \int d \ln k \mathcal{R}_{\text{MAP}}(k) W^2(kR). \quad (6)$$

This is the natural first thing to try, but it is not the natural object for a rotation-curve sign test. Equation (6) measures a smoothed amplitude. A rotation curve responds to a radial force gradient.

4.3 Why the bridge is derivative-like

For a circular velocity profile,

$$V_c^2(r) = \frac{GM(< r)}{r}. \quad (7)$$

At fixed observed radius, a small halo response changes the velocity through the radial mass profile rather than through the total smoothed mass alone:

$$\delta \ln V_c^2(r) \simeq \delta \ln M(< r). \quad (8)$$

The sign of the required correction therefore depends on how the response changes the outer profile, not merely on whether the total halo amplitude is increased or decreased. In a windowed

description, the lowest-order conversion of a mass-amplitude response into a radial-response proxy is a logarithmic derivative,

$$\Delta_{\nabla}(R) \equiv -\frac{\partial \Delta_{W^2}(R)}{\partial \ln R} = -\int d \ln k \mathcal{R}_{\text{MAP}}(k) \frac{\partial W^2(kR)}{\partial \ln R}. \quad (9)$$

Since $\partial/\partial \ln R = (kR)\partial/\partial(kR)$, Eq. (9) emphasizes the high- k shoulder and the slope of the response. Up to an overall sign convention and slowly varying factors, the leading positive proxy is

$$\Delta_{kW^2}(R) \propto \int d \ln k \mathcal{R}_{\text{MAP}}(k) k W^2(kR). \quad (10)$$

This is the origin of the kW^2 kernel used below. The k^2W^2 and band-limited W^2 kernels are not independent model freedoms; they are stress tests of the same hypothesis that SPARC couples to a derivative/high- k response rather than to the plain variance window.

4.4 Coupling to the measured outer slope

A response kernel alone cannot determine the observed sign without the state of the galaxy on which it acts. For the active SPARC audit, the relevant state variable is the measured outer slope m_{out} . The fixed pre-test prediction is therefore

$$\hat{y}_i = -\text{sign}[\Delta_{\nabla}(R_i) m_{\text{out},i}] \simeq -\text{sign}[\Delta_{kW^2}(R_i) m_{\text{out},i}]. \quad (11)$$

The minus sign is a convention inherited from the archived rotation-curve response: it maps a positive response acting on a rising or falling outer profile into the required half-space of A_{req} . Equation (11) is the rule tested in the rest of the paper. The SPARC analysis therefore tests a theory-motivated sign functional, not a free per-galaxy fit.

5 Data, target, and fixed rules

The data set is the active subset of SPARC galaxies selected in the existing SPARC audit pipeline. Each galaxy i has a required correction sign, encoded by $y_i = \text{sign}(A_{\text{req},i})$, and an observed outer-slope variable $m_{\text{out},i}$. The fixed no-fit rule tested for a response kernel K is

$$\hat{y}_i^{(K)} = -\text{sign}[\Delta_K(M_i) m_{\text{out},i}], \quad (12)$$

where $\Delta_K(M_i)$ is the MAP1782 halo-response proxy evaluated for the reference halo associated with galaxy i .

The sign score is

$$N_{\text{ok}}^{(K)} = \sum_i \mathbf{1}(\hat{y}_i^{(K)} = y_i), \quad (13)$$

with $N = 38$ active galaxies. The null model used for the headline significance is a target-sign permutation: the predicted signs are held fixed while the 38 target signs are randomly permuted. This tests whether the fixed rule aligns with the required SPARC half-space beyond the empirical target imbalance.

5.1 Operational active-sample definition

For v0.4 the active sample remains the same frozen operational set rather than being reselected in this manuscript. It is the 38-row table shipped as `sparc_kernel_variant_active_predictions.csv`, inherited from the SPARC audit branch. A galaxy enters this table only if all required quantities are finite in the frozen audit products: a SPARC rotation-curve model, a reference NFW halo, the required correction amplitude A_{req} , the outer-slope diagnostic m_{out} , and the response columns used

in the kernel comparison. The same 38 identifiers are used in every sign, null, and rotation-curve χ^2 table; no galaxy is added or removed after inspecting the MAP1782 kernel score.

The word “active” is therefore operational. It denotes galaxies for which the observed outer-disk state makes a signed halo-response correction meaningful in the archived SPARC machinery. The parent SPARC catalogue contains 175 disk galaxies; 137 are not used in this sign/null audit because the frozen Paper-II pipeline does not provide the complete finite tuple required here: reference NFW halo, required correction amplitude A_{req} , outer-slope diagnostic m_{out} , and all response-kernel columns. This is not yet a first-principles formation criterion, and it is not a claim that the other 137 galaxies are physically irrelevant. The claims of this paper are conditional on this frozen 38-galaxy active set. Appendix A gives the complete list.

6 Why the naive mass-variance bridge fails

The first transparent MAP1782 rebuild produced a mass-only response $\delta R_1(M)$ from the reconstructed $R_{\text{BPP}}(k)$ spectrum. That object is not sufficient. In the active table, the rebuilt δR_1 is negative for all 38 galaxies, so the product rule $-\text{sign}(\delta R_1 m_{\text{out}})$ becomes exactly identical to $\text{sign}(m_{\text{out}})$.

This failure is not cured by a simple normalization, sign flip, or window change. Table 1 gives the decisive forensic result: the archived response has three positive active cases, while the mass-only rebuild misses all three. These galaxies are precisely the cases that separate a morphology-only rule from the stronger coupled signal.

Table 1: Three positive active cases in the archived response and their values in the naive MAP1782 mass-only rebuild. The mass-only rebuild does not recover the positive tail.

Galaxy	A_{req}	m_{out}	archived dR_1	rebuilt δR_1	$\log M_{200}$
F571-8	-2	4.8660	0.002686	-0.048252	12.05
ESO079-G014	-2	9.4086	0.023492	-0.049106	12.40
ESO563-G021	4	-14.2905	0.019122	-0.057304	13.47

The empirical conclusion is important for the theoretical interpretation: the SPARC bridge is not a simple smoothed mass-variance response. It requires a response sensitive to scale gradients or outer-halo structure.

7 Kernel dictionary used in the audit

Section 4 motivates the derivative response before the data test. The numerical audit then instantiates that response with four frozen kernel columns built from the same MAP1782 spectral input:

$$\Delta_{W^2} \quad \text{plain mass-variance smoothing,} \quad (14)$$

$$\Delta_{kW^2} \quad \text{leading logarithmic-gradient proxy,} \quad (15)$$

$$\Delta_{k^2W^2} \quad \text{more aggressive high-}k \text{ stress test,} \quad (16)$$

$$\Delta_{\text{band}} \quad \text{finite-band response check.} \quad (17)$$

Only the first of these would be expected if SPARC were sensitive to integrated mass variance alone. The core falsifying check for the bridge is therefore simple: if the theory section is wrong, the W^2 row should work as well as, or better than, the derivative rows. The opposite is observed below.

8 Kernel forensic: from W^2 to kW^2

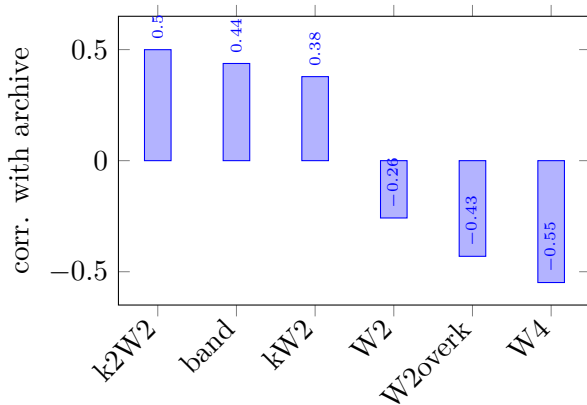
We therefore compare several kernels built from the same MAP1782 spectral information. The relevant modes are:

- W^2 : mass-variance-like smoothing;
- kW^2 : derivative-like or scale-gradient weighted response;
- k^2W^2 : more high- k weighted robustness variant;
- band-limited W^2 : finite k -window response.

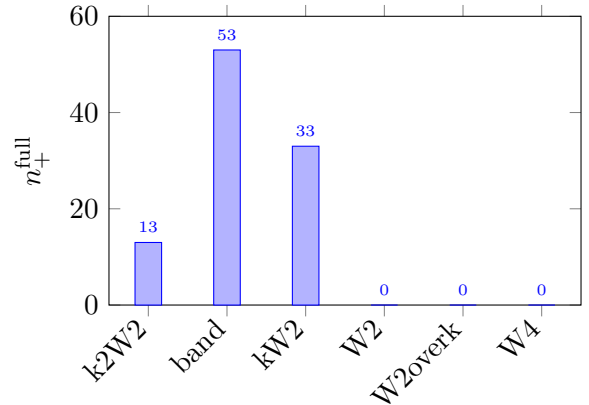
Table 2 summarizes the comparison to the archived R_1 response across the full available mass table, not only the 38 active galaxies. The mass-variance-like W^2 kernel has negative correlation with the archive and no positive tail. The kW^2 kernel has nearly perfect sign agreement with the archive response while recovering 33 positive cases in the high-mass range. The n_+ counts in Table 2 are full-table counts; they can therefore exceed the 38 active galaxies used in the sign test. The k^2W^2 kernel has higher linear correlation but is more selective and changes the active-sign performance slightly.

Table 2: Kernel forensic against the archived R_1 response. The W^2 row shows why the naive bridge fails; the kW^2 row is the leading bridge candidate. Here n_+^{full} is counted over the full response table, so the band value $n_+^{\text{full}} = 53$ is not in conflict with the 38-galaxy active sample of Table 3.

Kernel	corr. archive	sign agreement	n_+^{full}	positive log M range
k^2W^2	0.499621	0.897143	13	12.40–13.60
band 0.5–5 W^2	0.437497	0.874286	53	11.70–13.60
kW^2	0.378311	0.988571	33	12.00–13.60
W^2	-0.258071	0.822857	0	–
W^2/k	-0.430644	0.822857	0	–
W^4	-0.548689	0.822857	0	–



(a) Correlation with archived response.



(b) Recovered positive response tail.

Figure 1: Kernel forensic. A plain W^2 smoothing has no positive tail; derivative-like kernels recover it.

9 Fixed no-fit sign prediction

Table 3 gives the fixed sign-rule test on the 38 active SPARC galaxies. The headline rule is

$$\hat{y}_i = -\text{sign}\left(\Delta_{kW^2,i} m_{\text{out},i}\right). \quad (18)$$

It gives 33 correct signs out of 38, with $p_{\text{perm}} = 0.00014$ and $Z_{\text{perm}} = 4.059$ under the target-sign permutation null.

Table 3: Fixed no-fit sign rules on 38 active SPARC galaxies. Target: $\text{sign}(A_{\text{req}})$. Null: permutation of target signs.

Rule	Correct	Accuracy	p_{perm}	Z
$-\text{sign}(kW^2 m_{\text{out}})$	33/38	0.868421	0.00014	4.059
$-\text{sign}(k^2W^2 m_{\text{out}})$	32/38	0.842105	0.00094	3.633
$-\text{sign}(\text{band } m_{\text{out}})$	31/38	0.815789	0.00352	3.192
$-\text{sign}(W^2 m_{\text{out}})$	30/38	0.789474	0.01354	2.712
$\text{sign}(m_{\text{out}})$	30/38	0.789474	0.01434	2.699

The W^2 row is not an independent success: because $\Delta_{W^2} < 0$ for all active galaxies, it is effectively the m_{out} baseline. The gain from kW^2 is therefore the recovery of the extra response information that the mass-variance kernel erased.



Figure 2: Collapse-to-morphology check. On the 38 active galaxies, the W^2 product rule is identical to $\text{sign}(m_{\text{out}})$ in all 38 cases. The kW^2 rule differs in three cases, and those three differences are precisely the decisive positive-tail galaxies of Table 4.

10 The three decisive galaxies

The improvement from 30/38 to 33/38 is not a diffuse bookkeeping artefact. It is localized in three high-leverage active galaxies. Table 4 shows that the kW^2 response flips exactly the cases that the naive W^2 or δR_1 rebuild misses.

11 Rotation-curve $\Delta\chi^2$ validation

The sign test is necessary but not sufficient. The stronger check is to pass the selected half-space through the validated archived SPARC rotation-curve machinery. That machinery uses

$$V_{200}^{\text{BPB}} = V_{200}^{\text{ref}} (1 + \alpha \Delta), \quad (19)$$

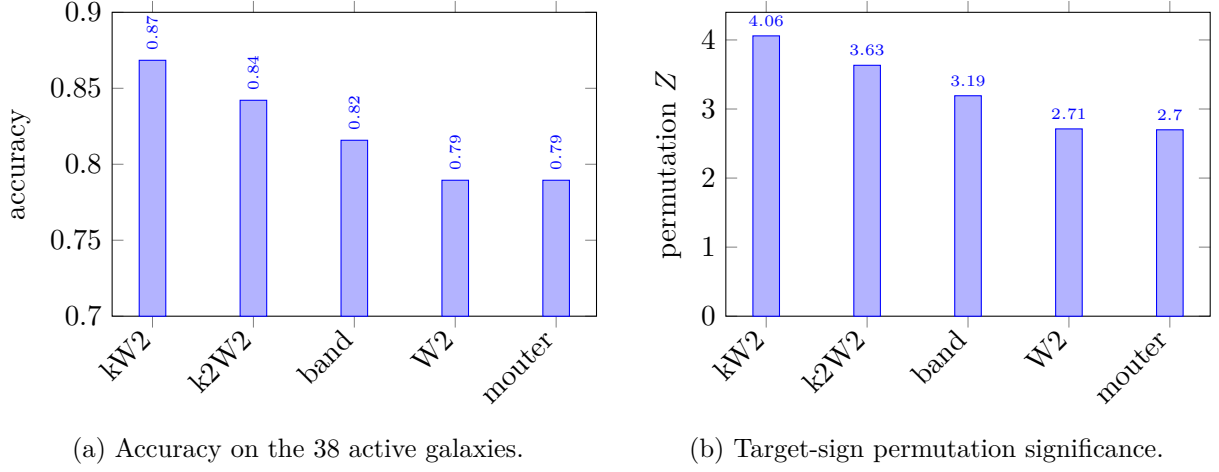


Figure 3: Fixed no-fit sign validation. The kW^2 rule is the leading candidate by both accuracy and permutation significance.

Table 4: Three decisive active galaxies. The kW^2 product rule is correct in all three cases, while the naive W^2 product is wrong.

Galaxy	A_{req}	m_{out}	δ_{W^2}	δ_{kW^2}	$\hat{y}_{W^2}$	$\hat{y}_{kW^2}$
F571-8	-2	4.8660	-0.174603	0.010375	+	-
ESO079-G014	-2	9.4086	-0.177693	0.069299	+	-
ESO563-G021	4	-14.2905	-0.207357	0.080191	-	+

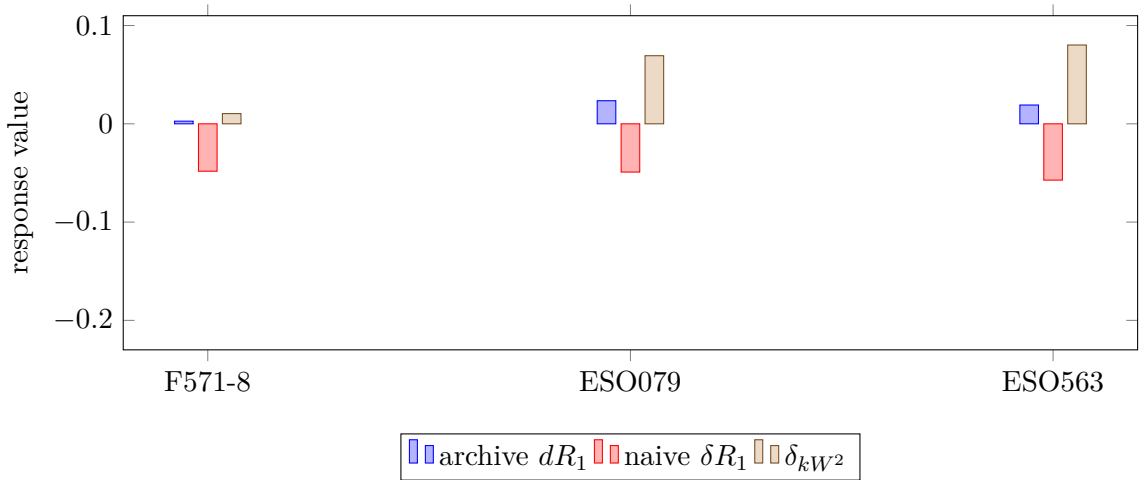


Figure 4: Positive-tail forensic for the decisive galaxies. The naive mass-only rebuild remains negative; kW^2 restores the sign structure needed by the product rule.

with the reference scale radius modified consistently by the same halo-ratio convention. The archive sanity row is reproduced: the archived 29/38 physical bridge gives $\Delta\chi^2 = -4107.590$, matching the recomputed -4107.581 within numerical roundoff.

Table 5 gives the resulting hierarchy. The MAP1782+ kW^2 product rule, evaluated with the validated archive amplitude calibration, gives $\Delta\chi^2 = -4605.230$, within exactly 8.340 of the archive oracle and exactly 497.640 better than the old 29/38 physical archive rule. The direct raw kW^2 amplitude row remains exploratory; it has the correct sign in 33/38 but is not yet fully amplitude-calibrated.

Table 5: Rotation-curve $\Delta\chi^2$ hierarchy on the 38 active galaxies.

Rule	Sign	$\sum \Delta\chi^2$
Archived oracle	38/38	-4613.570
Archive response oracle recomputation	38/38	-4613.564
MAP1782 kW^2 product + archive response calibration	33/38	-4605.230
Archived physical bridge	29/38	-4107.590
MAP1782 kW^2 product + raw kW^2 response	33/38	-3269.980
MAP1782 k^2W^2 product + raw k^2W^2 response	32/38	370.640

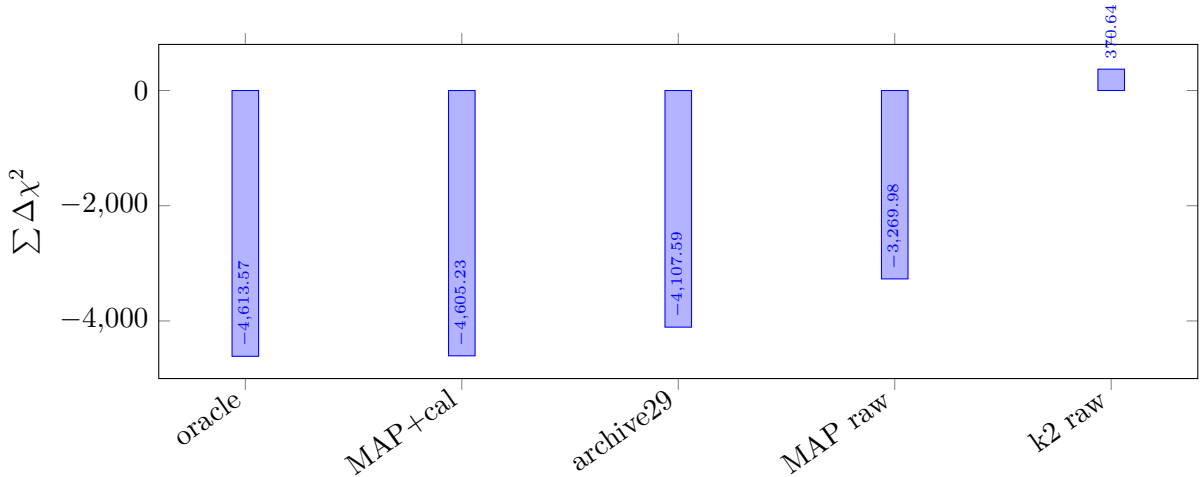


Figure 5: Rotation-curve $\Delta\chi^2$ hierarchy. The calibrated MAP1782+ kW^2 product rule is nearly oracle-level; the raw kW^2 amplitude is not yet the final calibrated amplitude model.

12 Self-calibrated response test

To remove dependence on the archived A_{req} amplitude, the self-calibrated diagnostic recomputes, for each response column, the best per-galaxy α directly from the rotation curves. The fixed product rule is then used only to select the sign half-space. For δ_{kW^2} , the product-gated result gives $\Delta\chi^2 = -4820.313$ compared with its self-calibrated oracle -4828.628 . The gap is only 8.315 in χ^2 . Across competent response columns the oracle-minus-gated gaps are remarkably uniform, 8.308–8.327, which indicates that the selection machinery is the fixed sign half-space rather than a special amplitude tuning of one kernel.

The δ_{W^2} row must be interpreted carefully. It is strong in this diagnostic because its response is negative for all active galaxies, so the product rule reduces to the outer-slope half-space. It does not solve the positive-tail problem and therefore cannot replace the kW^2 bridge candidate.

Table 6: Self-calibrated response test. For each response, the best per-galaxy amplitude is recomputed from the rotation curves.

Response	oracle self $\Delta\chi^2$	product sign	product-gated $\Delta\chi^2$	$n(\Delta > 0)$
δ_{W^2}	-4863.107	33/38	-4854.786	0
dR_1	-4836.615	33/38	-4828.292	3
δ_{kW^2}	-4828.628	33/38	-4820.313	3
$\delta_{k^2W^2}$	-4824.917	33/38	-4816.609	2
$\delta_{\text{band } W^2}$	-4091.578	33/38	-4083.251	5

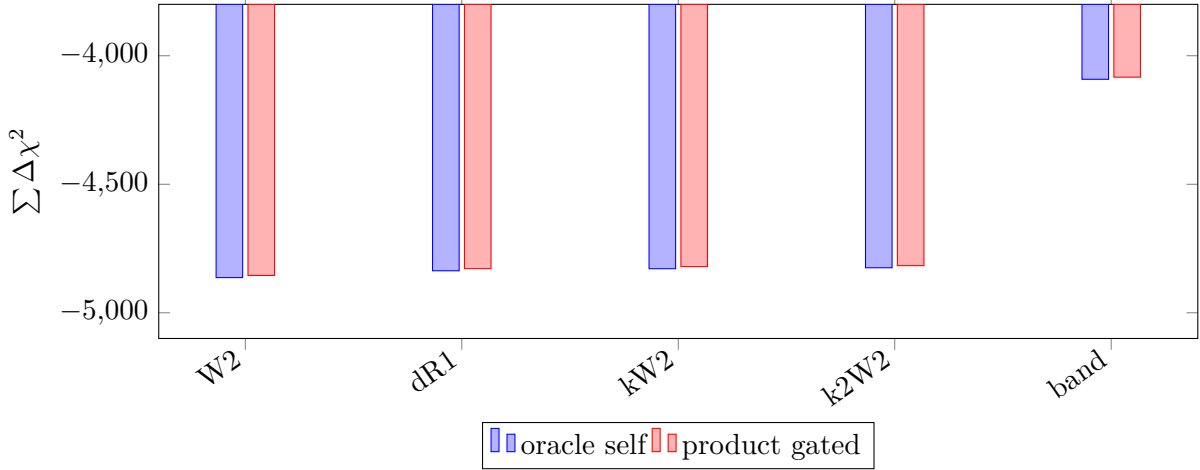


Figure 6: Self-calibrated rotation-curve test. For kW^2 , the fixed product rule selects a half-space within $\Delta\chi^2 \simeq 8.3$ of its self-calibrated oracle.

13 Amplitude proxy in A_{req} space

The continuous A_{req} -space proxy is not the true rotation-curve χ^2 , but it checks whether the same bridge carries amplitude information beyond the sign. Table 7 shows that $kW^2 \times m_{\text{out}}$ improves the A -space proxy relative to both m_{out} alone and $W^2 \times m_{\text{out}}$. The strongest raw correlation in this diagnostic is $k^2W^2 \times m_{\text{out}}$, but its direct rotation-curve amplitude is unstable in Table 5; hence it remains a robustness variant rather than the main bridge.

Table 7: Continuous amplitude proxy in A_{req} space. This is a proxy diagnostic, not the final rotation-curve likelihood.

Feature	Sign	$\text{corr}(A_{\text{req}}, A_{\text{pred}})$	χ_A^2 proxy	$\Delta\chi_A^2$ vs mean
$kW^2 \times m_{\text{out}}$	33/38	0.642071	54.778	-35.337
archive $dR_1 \times m_{\text{out}}$	33/38	0.538220	64.014	-26.100
$k^2W^2 \times m_{\text{out}}$	32/38	0.673996	51.930	-38.184
band $W^2 \times m_{\text{out}}$	31/38	0.614604	56.205	-33.910
m_{out} only	30/38	0.337610	81.133	-8.982
$W^2 \times m_{\text{out}}$	30/38	0.305282	83.654	-6.460

14 D5/D6 row-dependent amplitude lock

The sign result and the archived-response $\Delta\chi^2$ hierarchy show that the derivative half-space is real, but v0.3 still left a fair amplitude objection: the one-scale raw kW^2 amplitude did not recover the

full archive response. D5/D6 address precisely this point. They do not alter the 33/38 sign rule; they ask whether the remaining amplitude gap is row-dependent and whether a source-only kernel can recover most of it in raw point-level SPARC rotation curves.

D5-A5 first identifies the missing factor as row-dependent. The median archive-over-clean response scale is 0.933557277247, and the top-five residual drivers contain 80.7875% of the absolute gap. D5-A6 then fits a leave-one-out source-only kernel using halo mass and concentration-side predictors, with no archive or χ^2 target leakage. In compressed form it gives $\sum \Delta\chi^2 = -4309.449661$ and recovers 0.778483 of the clean-to-archive gap. D5-A6b verifies name recovery, leakage guards, bootstrap coefficient behavior, and repeated five-fold resampling, for which the median recovered fraction is 0.790380.

The decisive validation is D5-A7: the fixed A6 kernel is injected into the raw point-level SPARC rotation-curve calculation. The primary source-only row-dependent kernel gives

$$\sum \Delta\chi_{\text{raw}}^2 = -4302.455768, \quad f_{\text{rec}} = 0.773245, \quad \Delta\chi_{\text{vs archive}}^2 = 302.773824, \quad (20)$$

with no invalid ratio rows and no negative χ^2 rows. The c200-only source-side control is similarly strong, with $\sum \Delta\chi_{\text{raw}}^2 = -4308.429759$ and $f_{\text{rec}} = 0.777720$. The correct interpretation is that source-side concentration/halo-shape information carries most of the remaining amplitude gap; it is not an exact archive or oracle recovery.

D6 adds the falsification layer. With 5000 source-feature permutations, no permuted model reaches the real recovered fraction, giving the exact empirical lower-bound value $p = 1/(5000 + 1) = 0.000199960008$. The null median recovered fraction is negative, while the real value is 0.773245. Top-driver removal also keeps the effect visible: removing the largest absolute-gap driver gives $f_{\text{rec}} = 0.832030$, removing the top three gives 0.746563, removing the top five gives 0.580735, and removing the top ten gives 0.796819. These are robustness audits, not replacements for the full 38-galaxy result.

Table 8: D5/D6 raw amplitude-kernel lock. These rows update the v0.3 amplitude guardrail while retaining the no-exact-oracle guardrail.

Audit row	$\sum \Delta\chi_{\text{raw}}^2$	recovered gap fraction	note
A6 primary $\log M_{200} + \text{c200}$ kernel	-4302.456	0.773245	raw RC validated
A6 c200-only control	-4308.430	0.777720	source-side control
A6 compressed LOO source model	-4309.450	0.778483	pre-raw anchor
Repeated 5-fold median	-4325.335	0.790380	robustness
Archive response lock	-4605.230	1.000000	not exactly recovered
Oracle archive response	-4613.564	1.006242	unattained oracle

This section changes the Paper-II status. The safe v0.4 claim is no longer only a sign bridge plus archived amplitude check. It is now a raw-rotation-curve validated, source-only row-dependent amplitude closure for most of the clean-to-archive gap, with permutation, top-driver, and ablation guardrails. The forbidden claim remains exact archive/oracle recovery and a full nonlinear galaxy-formation derivation of the kernel.

15 Interpretation

The present result has a precise interpretation. The MAP1782 SPARC bridge is not a statement that every galaxy receives a universal scalar boost. It is a statement that a particular spectral response, when filtered with a derivative-like kernel and combined with the observed outer rotation-curve state, selects the correct correction half-space in the active SPARC sample.

This mirrors the lesson of Paper I. In Paper I, μ_m and Σ are not the same observable window. Here, the mass-variance-like W^2 window and the derivative-like kW^2 window are not equivalent.

A simple smoothed response erases the high-mass positive tail; the derivative-like response restores it.

The three decisive galaxies make the point concrete. In F571-8, ESO079-G014, and ESO563-G021, the naive rebuilt δR_1 remains negative, while δ_{kW^2} becomes positive. The sign of m_{out} then produces the correct required correction in all three cases. This is the difference between the 30/38 morphology-only baseline and the 33/38 MAP1782+ kW^2 rule.

This interpretation is deliberately distinct from a MOND or RAR claim. Milgrom’s original proposal [5] and the observed radial-acceleration relation [6] are relevant comparison points for any rotation-curve paper, but the present test is not a fit of a MOND interpolation function and does not infer a universal acceleration scale from the SPARC curves. The Paper-II object is a frozen cosmological-response half-space, filtered through a derivative-like halo kernel; the Paper-VII BTFR/RAR-style acceleration tests are kept separate in the corpus.

16 Claim boundaries

The claims supported by this v0.4 draft, with the theory-first ordering of Section 4, are:

1. MAP1782+ $kW^2 \times m_{\text{out}}$ gives 33/38 correct active-SPARC signs with $p_{\text{perm}} = 0.00014$.
2. The naive W^2 bridge fails as an independent SPARC response because it has no positive active tail and collapses to the morphology-only baseline.
3. The same 33/38 rule, inserted into the validated archived rotation-curve machinery, gives $\Delta\chi^2 = -4605.230$, nearly the archived oracle value -4613.570 .
4. A self-calibrated rotation-curve test confirms that the result does not rely only on the archived A_{req} amplitudes: δ_{kW^2} gives $\Delta\chi^2 = -4820.313$ under product-gating versus -4828.628 for its own oracle.
5. D5/D6 promote the amplitude statement to a raw point-level validation: a source-only row-dependent kernel gives $\sum \Delta\chi_{\text{raw}}^2 = -4302.455768$ and recovers 0.773245 of the clean-to-archive gap, with a 5000-draw source-feature permutation lower bound $p = 1/5001$.
6. D3 closes the projected Weyl-matter normalization used in the wider Paper-II/Paper-VI chain as $r_{W\delta} = \sqrt{a_{\text{bg}}} = 0.635364421415$, with an s_0 -weighted sub-percent mechanism diagnostic.
7. D7 validates the face-minus $\Delta n_s = 0.010$ mechanism at CAMB spectra level, with TT/EE effective slopes 0.008945 and 0.009453 over $\ell = 600\text{--}2200$.

The claims not made are equally important:

1. We do not claim that BPB/MBE predicts m_{out} from cosmology alone in this draft.
2. We do not claim that the one-scale raw kW^2 amplitude is fully calibrated. The v0.4 amplitude claim is instead the D5/D6 row-dependent, source-only kernel validated at raw RC level.
3. We do not claim exact archive or oracle recovery. The primary D5 raw kernel remains $\Delta\chi^2 = 302.773824$ away from the archive response and 311.108489 away from the oracle.
4. We do not claim that kW^2 is already derived as the unique fundamental kernel. At v0.4 it is the leading theory-motivated proxy for a derivative or outer-response halo calculation; the full nonlinear derivation is deferred.
5. We do not claim a SPARC-only discovery. The sample has 38 active galaxies, and the result is best understood as an external validation of the independently frozen Paper-I branch.
6. We do not claim a full Planck likelihood from D7, nor a full SPARC galaxy-level nuisance/covariance likelihood from D5/D6.

17 Conclusions

1. Paper I fixes MAP1782 independently of SPARC. Paper II uses SPARC as an external bridge validation target and records its corpus dependencies through the root release index.
2. D3 closes the projected Weyl-matter normalization used by the Paper-II/Paper-VI chain as $r_{W\delta} = \sqrt{a_{\text{bg}}} = 0.635364421415$, with the s_0 -weighted projector recorded as a sub-percent but non-exact mechanism diagnostic.
3. The transparent mass-only MAP1782 rebuild fails in the active sample: δR_1 is negative for all 38 active galaxies and the product rule becomes identical to $\text{sign}(m_{\text{out}})$.
4. Kernel forensic identifies a derivative-like response as the relevant object. The kW^2 kernel recovers the high-mass positive response tail missed by W^2 .
5. The fixed rule $-\text{sign}(\delta_{kW^2} m_{\text{out}})$ gives 33/38 correct signs, accuracy 0.868421, $p_{\text{perm}} = 0.00014$, and $Z = 4.059$.
6. The gain is localized in three decisive galaxies: F571-8, ESO079-G014, and ESO563-G021. These are exactly the active positive-tail cases missed by the naive mass-only rebuild.
7. Rotation-curve validation gives $\Delta\chi^2 = -4605.230$ for the calibrated MAP1782+ kW^2 product rule, compared with -4107.590 for the old archived physical rule and -4613.570 for the archived oracle.
8. Self-calibrated tests confirm that the kW^2 half-space remains near-oracle when the per-galaxy amplitude is recomputed from rotation curves.
9. D5/D6 add the v0.4 amplitude closure: a source-only row-dependent kernel recovers most of the clean-to-archive amplitude gap directly in raw point-level SPARC rotation curves, with $\sum \Delta\chi_{\text{raw}}^2 = -4302.455768$, recovered fraction 0.773245, no bad rows, a strong c200-only control, and a 5000-draw source-feature permutation lower-bound $p = 1/5001$.
10. D7 verifies the associated face-minus tilt mechanism in CAMB but remains a mechanism gate, not a full Planck likelihood.
11. The remaining SPARC work is the full nonlinear formalization of the kW^2 response kernel and, separately, a true galaxy-level nuisance/covariance likelihood. These are not hidden; they are the explicit next theoretical and statistical extensions.

A Frozen active-galaxy list

The active sample used throughout the sign, null, and rotation-curve tests consists of the following 38 galaxies. This list is frozen before the kernel comparison and is not reselected by the kW^2 score.

UGCA444	KK98-251	DDO064	UGCA442
DDO168	D631-7	UGC01281	UGC07603
UGC05721	NGC2915	DDO161	UGC08286
UGC06818	UGC04499	F571-V1	F583-1
NGC3109	UGC06399	UGC05005	NGC0300
NGC2976	UGC04278	IC2574	NGC0100
NGC5585	UGC00128	ESO116-G012	NGC4085
NGC3972	NGC1003	NGC4010	NGC3877
UGC05986	NGC1090	NGC3917	F571-8
ESO079-G014	ESO563-G021		

B Data and reproducibility inventory

The numerical statements in this v0.4 source are tied to the corpus release index and the following source-lock artefacts:

- 00_README_CORPUS, 00_MANIFEST_CORPUS, 00_CLAIMS_INDEX, and 00_REVIEWER_ROUTE: root corpus navigation and claim index;
- outputs/paperII_D3/paperII_D3_projector_closure_lock.md/json/csv: D3 density-weighted projector lock;
- outputs/paperII_D3/paperII_D3b_s0_weighted_projector_audit.md/json/csv: D3b s_0 -weighted sub-percent projector diagnostic;
- sparc_final_closure_summary.md/json: headline v0.3 closure table;
- sparc_kernel_product_nulls.csv/md: fixed sign rules and target-sign permutation nulls;
- sparc_kernel_variants_vs_archive.csv/md: kernel forensic against archived R_1 ;
- sparc_dR1_positive_tail_forensic.csv/md: three-galaxy positive-tail diagnostic;
- sparc_kernel_variant_active_predictions.csv: active-sample per-galaxy sign predictions;
- sparc_kW2_rotation_chi2_summary.csv/md: archived-response rotation-curve $\Delta\chi^2$ hierarchy;
- sparc_kW2_selfcalibrated_chi2_summary.csv/md: self-calibrated response test;
- sparc_kW2_amplitude_proxy.csv/md: continuous A_{req} proxy diagnostic;
- outputs/paperII_D5/paperII_D5A8_final_D5_lock.md/json/csv: D5 raw-RC amplitude-kernel lock;
- outputs/paperII_D5/paperII_D6_kernel_falsification_lock.md/json/csv: D6 permutation, top-driver, ablation, and final amplitude-kernel falsification lock;
- outputs/paperII_D7/paperII_D7_camb_face_minus_gate.md/json/csv: D7 CAMB face-minus tilt mechanism gate.

C Version notes

This is v0.4.1, the corpus-aware Paper-II source hotfix. Relative to v0.3, it keeps the original theory-first ordering and all original SPARC sign/rotation-curve numbers, but adds the post-v0.3 source locks that closed the remaining Paper-II dashboard items:

- D3/D3b: density-weighted projector closure, $r_{W\delta} = \sqrt{a_{\text{bg}}} = 0.635364421415$, with an s_0 -weighted sub-percent but non-exact mechanism diagnostic;
- D5: raw point-level SPARC validation of the row-dependent source-only amplitude kernel, with primary $\sum \Delta\chi_{\text{raw}}^2 = -4302.455768$ and recovered fraction 0.773245;
- D6: permutation/null/top-driver/ablation falsification of the D5 kernel, with empirical $p = 1/5001$ for the 5000-draw source-feature permutation audit;
- D7: CAMB face-minus tilt gate, with $\Delta n_s = 0.010$ and TT/EE slopes 0.008945 and 0.009453 over $\ell = 600\text{--}2200$.

The original v0.3 numerical sign results are unchanged: 33/38, $p_{\text{perm}} = 0.00014$, $Z = 4.059$, calibrated archived-response rotation-curve $\Delta\chi^2 = -4605.230$, oracle gap 8.340, and old-archive improvement 497.640. The v0.4 changes were release navigation, explicit D3/D7 corpus placement, and the D5/D6 amplitude-kernel lock; v0.4.1 adds the active-sample exclusion count, labels the reproducibility appendix, standardizes the permutation notation, and adds MOND/RAR comparison references without changing any scientific number.

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